

Unit 9 Progress Check: MCQ

Table 1

Cell Type	Operating Cell Potential for Commercial Batteries, E (V)
Lithium-iodine	+2.80
Zinc-mercury	+1.35

Table 2

Half-Reaction	Standard Reduction Potential, E° (V)
$[\text{Zn}(\text{OH})_4]^{2-} + 2 e^- \rightarrow \text{Zn} + 4 \text{OH}^-$	-1.20
$\text{Zn}(\text{OH})_2 + 2 e^- \rightarrow \text{Zn} + 2 \text{OH}^-$	-1.25
$\text{HgO} + \text{H}_2\text{O} + 2 e^- \rightarrow \text{Hg} + 2 \text{OH}^-$	+0.10
$\text{O}_2 + 2 \text{H}_2\text{O} + 4 e^- \rightarrow 4 \text{OH}^-$	+0.40

Pacemakers are electronic devices that help regulate the heart rate. Currently, lithium-iodine cells are commonly used to power pacemakers and have replaced zinc-mercury cells. Table 1 provides the operating cell potential, E , for each cell. Table 2 provides the standard reduction potentials for several half-reactions related to zinc-mercury and zinc-air cells.

1. Based on the information given, which of the following is a major difference between the zinc-mercury cell and the lithium-iodine cell?
- (A) During the initial cell operation, each reaction is thermodynamically favorable, but the larger operating potential of the lithium-iodine cell indicates that its cell reaction is less thermodynamically favorable.
- (B) During the initial cell operation, each reaction is thermodynamically favorable, but the larger operating potential of the lithium-iodine cell indicates that its cell reaction is more thermodynamically favorable. ✓
- (C) During the initial cell operation, the oxidation of iodine is thermodynamically favorable but the oxidation of mercury is not.
- (D) During the initial cell operation, the oxidation of mercury is thermodynamically favorable but not the oxidation of iodine is not.



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2. On average, after one year of operation, the potential of a lithium-iodine cell decreases by 1%-2%. Which of the following best helps to explain the cause for the decrease in cell potential?

- (A) $E^\circ_{\text{cell}} = E^\circ_{\text{red}}(\text{cathode}) - E^\circ_{\text{red}}(\text{anode})$, and as the cell operates, $E^\circ_{\text{red}}(\text{cathode})$ decreases.
- (B) $E^\circ_{\text{cell}} = E^\circ_{\text{red}}(\text{cathode}) - E^\circ_{\text{red}}(\text{anode})$, and as the cell operates, $E^\circ_{\text{red}}(\text{anode})$ increases.
- (C) $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$, and as the cell operates, Q decreases.
- (D) $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$, and as the cell operates, Q increases. ✓

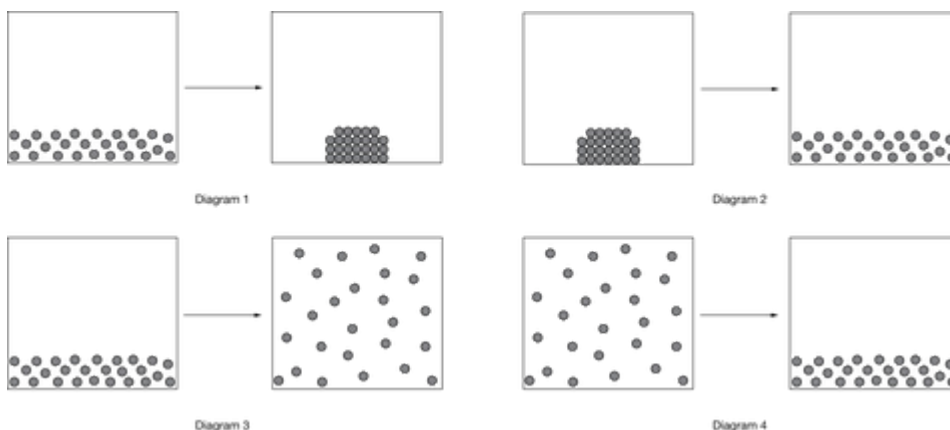
3. The use of zinc-mercury cells in hearing aids has been replaced by zinc-air cells that operate using the oxidation of **Zn** by **O₂** from the air, generating a potential of +1.60 V. Table 2 provides the standard reduction potentials for the half-reactions used in zinc-mercury and zinc-air cells. Which of the following best explains the modification to the cell design that is mostly responsible for the difference in standard cell potentials for zinc-mercury and zinc-air cells?

- (A) The greater standard cell potential of the **Zn-air** cell compared to that of the zinc-mercury cell most likely results from the thermodynamically more favorable reduction of **O₂** compared to **HgO**. ✓
- (B) The greater standard cell potential of the **Zn-air** cell compared to that of the zinc-mercury cell most likely results from the greater number of moles of e^- required to reduce **O₂** compared to **HgO**.
- (C) The greater standard cell potential of the **Zn-air** cell compared to that of the zinc-mercury cell most likely results from the thermodynamically less favorable reduction of **O₂** compared to **HgO**.
- (D) The greater standard cell potential of the **Zn-air** cell compared to that of the zinc-mercury cell most likely results from the greater number of moles of hydroxide ions required to reduce $[\text{Zn}(\text{OH})_4]^{2-}$ compared to **Zn(OH)₂**.



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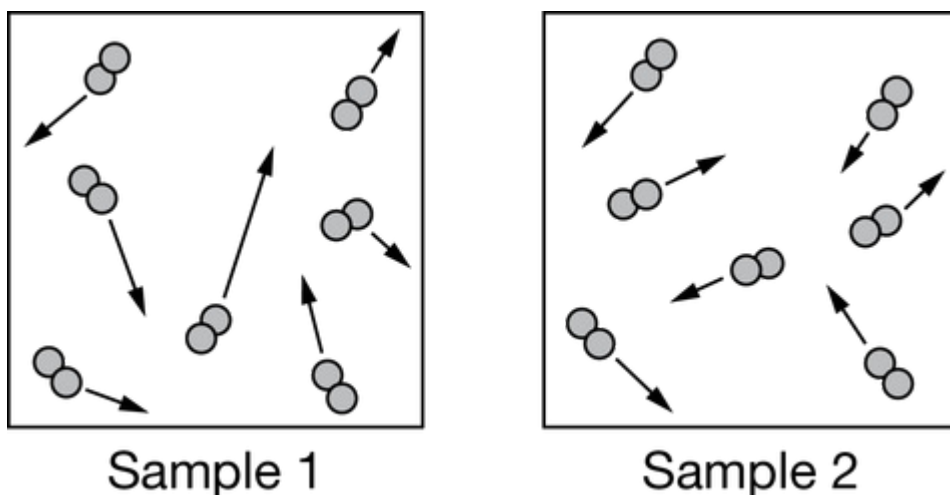
4.



The diagrams above represent physical changes for potassium. Which of the following correctly identifies the physical process and provides the correct ΔS° for the process in the indicated diagram?

- (A) In diagram 1, the process is freezing and $\Delta S^\circ_{\text{freezing}}$ is negative. ✓
- (B) In diagram 2, the process is melting and $\Delta S^\circ_{\text{melting}}$ is negative.
- (C) In diagram 3, the process is boiling and $\Delta S^\circ_{\text{boiling}}$ is negative.
- (D) In diagram 4, the process is condensation and $\Delta S^\circ_{\text{condensation}}$ is positive.

5.



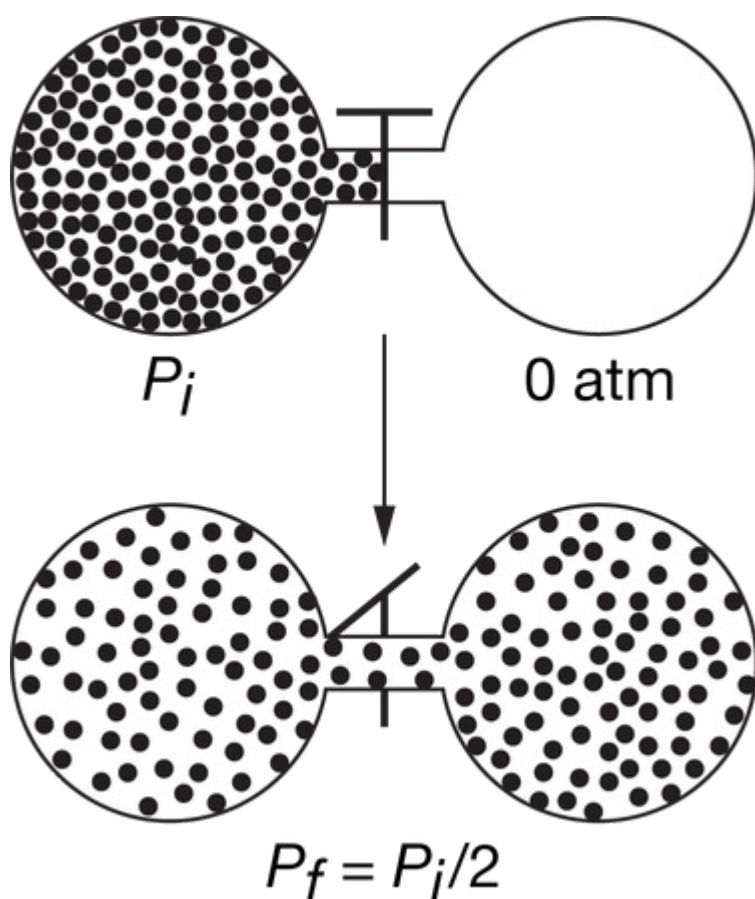
Two samples containing an equal number of moles of $\text{N}_2(g)$ are kept inside separate 1-liter rigid containers. The particle diagrams above show the distribution of molecular speeds in each sample. Based on the information given, which of the following identifies the sample with the greater S° , and why?



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- (A) Sample 1, because the N_2 molecules collide less frequently compared to sample 2.
- (B) Sample 1, because the N_2 molecules have a larger distribution of energies compared to sample 2. ✓
- (C) Sample 2, because the N_2 molecules collide more frequently compared to sample 1.
- (D) Sample 2, because the N_2 molecules have a smaller distribution of energies compared to sample 1.

6.



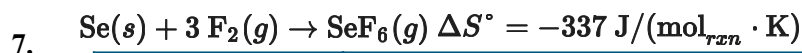
A sample of an ideal gas at P_i is initially confined to one chamber of the apparatus represented above, and the other chamber is initially evacuated. The valve connecting the chambers is opened, and the gas expands at constant temperature to fill both chambers. Which of the following best describes ΔS for the process?



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- (A) $\Delta S < 0$ because the pressure of the gas decreases by half.
- (B) $\Delta S = 0$ because there is no change in the total number of particles of the gas.
- (C) $\Delta S = 0$ because the average kinetic energy of the gas particles is unchanged.

(D) $\Delta S > 0$ because the gas particles become dispersed in a larger volume. ✓



Substance	Approximate S° (J/(mol · K))
Se(s)	42
F ₂ (g)	?
SeF ₆ (g)	314

The chemical equation above represents the formation of $\text{SeF}_6(g)$ from its elements and the table provides the approximate values of S° for $\text{Se}(s)$ and $\text{SeF}_6(g)$. Based on the data, which of the following mathematical expressions can be used to correctly calculate S° for $\text{F}_2(g)$?

- (A) $S^\circ = [-337 + 314 - 42] \text{ J}/(\text{mol} \cdot \text{K})$
- (B) $S^\circ = -[-337 - 314 + 42] \text{ J}/(\text{mol} \cdot \text{K})$
- (C) $S^\circ = \frac{1}{3}[-337 + 314 - 42] \text{ J}/(\text{mol} \cdot \text{K})$

(D) $S^\circ = -\frac{1}{3}[-337 - 314 + 42] \text{ J}/(\text{mol} \cdot \text{K})$ ✓

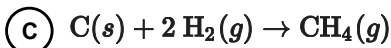
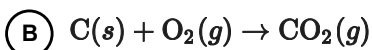
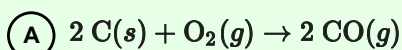


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8.

Substance	Approximate S° (J/(mol · K))
C(s)	6
CO(g)	198
CO ₂ (g)	214
CH ₄ (g)	186
C ₂ H ₆ (g)	229
H ₂ (g)	131
O ₂ (g)	205

The table above provides approximate S° values for several substances. Based on the information, which of the following reactions has the largest increase in entropy, ΔS° ?



Substance	Approximate S° (J/(mol · K))
C ₆ H ₁₂ O ₆ (s)	209
O ₂ (g)	205
H ₂ O(g)	?
CO ₂ (g)	214

The chemical equation above represents the combustion of glucose, and the table provides the approximate standard absolute entropies, S° , for some substances. Based on the information given, which of these equations can be used to calculate an approximation of S° for H₂O(g)?



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(A) $S^\circ = [900 + 209 + 205 - 214] \text{ J}/(\text{mol} \cdot \text{K})$

(B) $S^\circ = \frac{1}{6} [900 + 209 + (6 \times 205) - (6 \times 214)] \text{ J}/(\text{mol} \cdot \text{K})$ ✓

(C) $S^\circ = \frac{1}{6} [-900 - (6 \times 214) + 209 + (6 \times 205)] \text{ J}/(\text{mol} \cdot \text{K})$

(D) $S^\circ = [-900 - 214 + 209 + 205] \text{ J}/(\text{mol} \cdot \text{K})$

10. An ice cube at 0°C melts when placed inside a room at 22°C (295 K). Based on the concepts of enthalpy, entropy, and Gibbs free energy, which of the following best explains why the process is thermodynamically favorable?

(A) Melting ice releases energy, $\Delta H^\circ < 0$, and $\Delta S^\circ < 0$ because the motion of the H_2O molecules decreases as it transitions from solid to liquid. At a temperature higher than 0°C (273 K), the term $T\Delta S^\circ$ is smaller than ΔH° , resulting in a thermodynamically favorable process with $\Delta G^\circ < 0$.

(B) Melting ice releases energy, $\Delta H^\circ < 0$, and $\Delta S^\circ > 0$ because the motion of the H_2O molecules increases as it transitions from solid to liquid. At a temperature higher than 0°C (273 K), the term $T\Delta S^\circ$ is greater than ΔH° , resulting in a thermodynamically favorable process with $\Delta G^\circ > 0$.

(C) Melting ice requires energy, $\Delta H^\circ > 0$, and $\Delta S^\circ < 0$ because the motion of the H_2O molecules decreases as it transitions from solid to liquid. At a temperature higher than 0°C (273 K), the term $T\Delta S^\circ$ is smaller than ΔH° , resulting in a thermodynamically favorable process with $\Delta G^\circ > 0$.

(D) Melting ice requires energy, $\Delta H^\circ > 0$, and $\Delta S^\circ > 0$ because the motion of the H_2O molecules increases as it transitions from solid to liquid. At a temperature higher than 0°C (273 K), the term $T\Delta S^\circ$ is greater than ΔH° , resulting in a thermodynamically favorable process with $\Delta G^\circ < 0$. ✓

11. A 5.00 g -sample of $\text{KOH}(s)$ at 25.0°C was added to 100.0 g of $\text{H}_2\text{O}(l)$ at room temperature inside an insulated cup calorimeter, and the contents were stirred. After all the $\text{KOH}(s)$ dissolved, the temperature of the solution had increased. Based on the information given, which of the following best justifies the claim that the dissolution of $\text{KOH}(s)$ is a thermodynamically favorable process?



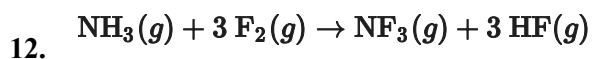
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- (A) The forces between the ions and the water molecules are stronger than the forces between water molecules, thus $\Delta H < 0$. Also, the ions become less dispersed as $\text{KOH}(s)$ dissolves, thus $\Delta S > 0$. Therefore, $\Delta G < 0$.

- (B) The energy required to break the bonds between the ions in the solid is less than that released as the ion-dipole attractions form during solvation, thus $\Delta H < 0$. Also, the ions become more widely dispersed as $\text{KOH}(s)$ dissolves, thus $\Delta S > 0$. Therefore, $\Delta G < 0$. ✓

- (C) The average kinetic energy of the particles increases, resulting in $\Delta H > 0$. Also, the ions become more widely dispersed as $\text{KOH}(s)$ dissolves, thus $\Delta S > 0$. Therefore, $\Delta G > 0$.

- (D) The average kinetic energy of the particles increases, resulting in $\Delta H > 0$. Also, the ions become more widely dispersed as $\text{KOH}(s)$ dissolves, thus $\Delta S < 0$. Therefore, $\Delta G > 0$.



The fluorination of ammonia is represented by the balanced equation above. Approximate values of ΔG°_f for the reactants and products are given in the following table.

Compound	ΔG°_f (kJ/mol)
$\text{NH}_3(g)$	−16
$\text{F}_2(g)$	0
$\text{NF}_3(g)$	−91
$\text{HF}(g)$	−280

Based on the information, which of the following statements best helps to explain whether or not the reaction is thermodynamically favored at 298 K?

- (A) $\Delta G^\circ_{rxn} \ll 0$ and the reaction is thermodynamically favored because the product molecules have more complex structures and greater absolute entropies than the reactant molecules do.

- (B) $\Delta G^\circ_{rxn} \gg 0$ and the reaction is not thermodynamically favored because the total number of moles of gas-phase products is not greater than the total number of moles of gas-phase reactants.

- (C) $\Delta G^\circ_{rxn} \ll 0$ and the reaction is thermodynamically favored because the energy released when the bonds in the products are formed is greater than the energy absorbed to break the bonds in the reactants. ✓

- (D) $\Delta G^\circ_{rxn} \gg 0$ and the reaction is not thermodynamically favored because the energy released when the bonds in the products are formed is less than the energy absorbed to break the bonds in reactants.

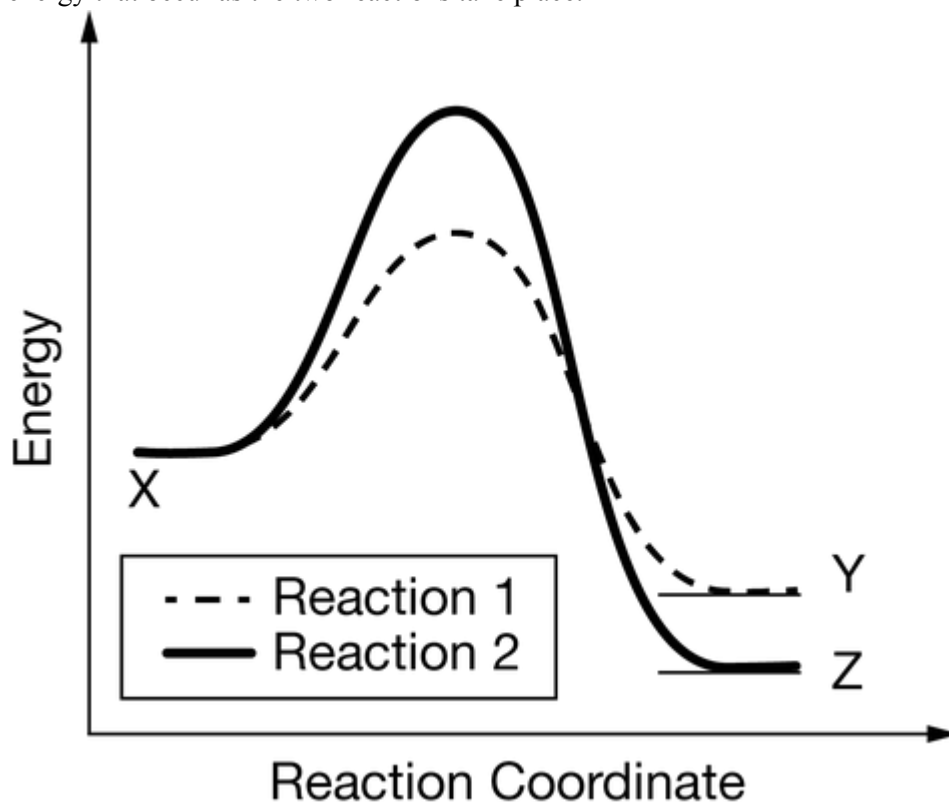


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13.

Reaction 1:	$X \rightleftharpoons Y$
Reaction 2:	$X \rightleftharpoons Z$

X undergoes the two competing reactions represented above. The following graph shows the changes in energy that occur as the two reactions take place.



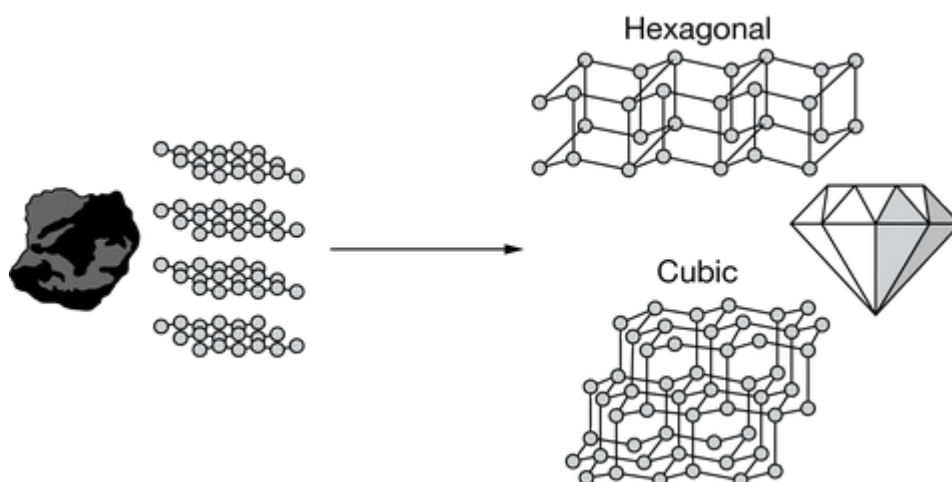
In an experiment done at a low temperature, one hour after the reaction started, **Y** was produced but almost no **Z** was produced. At a much higher temperature, one hour after the reaction started, both **Y** and **Z** had been produced. Which of the following best explains why very little **Z** was produced at the lower temperature?

- (A) A smaller number of **X** molecules had enough energy to overcome the activation energy barrier for reaction 2, and very little **Z** was produced. ✓
- (B) The higher average kinetic energy of the **X** molecules at the lower temperature resulted in very little **Z** produced.
- (C) The greater collision frequency between **X** molecules at the lower temperature resulted in very little **Z** produced.
- (D) The intermolecular forces between **X** molecules were much weaker, and very little **Z** was produced.



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14.



Graphite, an allotrope of carbon, is converted into cubic diamond through a process that may take a billion years or longer. As illustrated above, scientists can make synthetic diamonds using a certain process in about one week. However, these synthetic diamonds have carbon atoms in a hexagonal lattice. Diamonds with a carbon atoms in a cubic lattice are not produced even though they are thermodynamically more stable than hexagonal diamond. Which of the following best justifies why the synthetic process produces hexagonal diamond and not the more thermodynamically stable cubic diamond?

- (A) The amount of energy required to create new bonds between carbon atoms in cubic diamond is much greater than the amount of energy required to create hexagonal diamond.
- (B) The amount of energy required to create new bonds between carbon atoms in cubic diamond is much smaller than the amount of energy required to create hexagonal diamond.
- (C) The activation energy needed to form cubic diamond is much less than the activation energy needed to form hexagonal diamond.
- (D) The activation energy needed to form cubic diamond is much greater than the activation energy needed to form hexagonal diamond. ✓

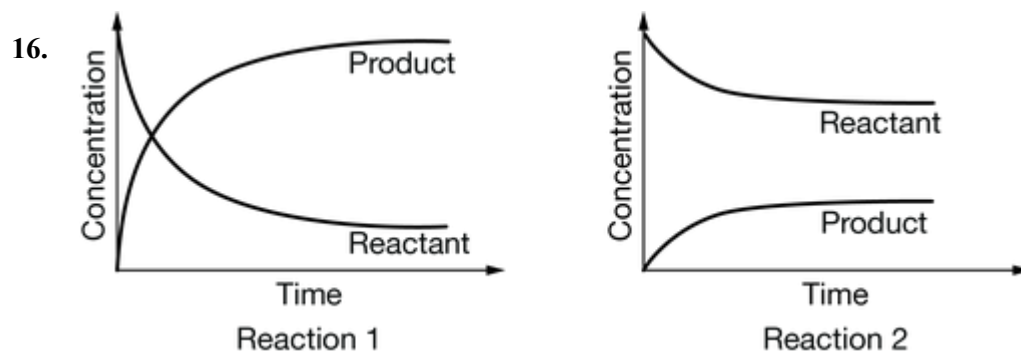
15. $\text{ATP} + \text{H}_2\text{O} \rightleftharpoons \text{ADP} + \text{P}_i \quad \Delta G = -30.5 \text{ kJ/mol}_{rxn}$

The hydrolysis of adenosine triphosphate (ATP) is represented by the equation above. This reaction is critically important in cellular biology, but the reaction itself proceeds at a very slow rate. Based on the information given, which of the following best explains why an enzyme (biological catalyst) is required for the reaction to occur at a faster rate?



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- (A) Because $\Delta G < 0$, the hydrolysis of **ATP** is not thermodynamically favorable. In cells, ΔS is increased by increasing the amount of **H₂O** consumed, resulting in $\Delta G > 0$ and an increase in the reaction rate.
- (B) Because $\Delta G < 0$, the hydrolysis of **ATP** is not thermodynamically favorable. In cells, enzymes act as catalysts that decrease ΔH for the reaction, resulting in $\Delta G > 0$ and an increase in the reaction rate.
- (C) Although the hydrolysis of **ATP** is thermodynamically favorable, without a catalyst the reaction occurs at a very slow rate because it has a small activation energy.
- (D) Although the hydrolysis of **ATP** is thermodynamically favorable, without a catalyst the reaction occurs at a very slow rate because it has a large activation energy. ✓

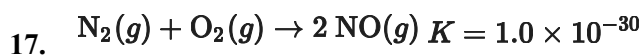


The graphs above represent the changes in concentration over time for the reactant and product of two separate reactions at 25°C . Based on these graphs, which of the following best supports the claim that reaction 1 is thermodynamically favorable but reaction 2 is not?

- (A) At equilibrium, for reaction 1 $\frac{[\text{Product}]}{[\text{Reactant}]} \gg 1$ and $\Delta G^{\circ}_{rxn} < 0$ but for reaction 2 $\frac{[\text{Product}]}{[\text{Reactant}]} \ll 1$ and $\Delta G^{\circ}_{rxn} > 0$. ✓
- (B) At equilibrium, for reaction 1 $\frac{[\text{Product}]}{[\text{Reactant}]} \gg 1$ and $\Delta G^{\circ}_{rxn} > 0$ but for reaction 2 $\frac{[\text{Product}]}{[\text{Reactant}]} \ll 1$ and $\Delta G^{\circ}_{rxn} < 0$.
- (C) At equilibrium, for reaction 1 $\frac{[\text{Product}]}{[\text{Reactant}]} = 1$ and $\Delta G^{\circ}_{rxn} = 0$ but for reaction 2 $\frac{[\text{Product}]}{[\text{Reactant}]} \ll 1$ and $\Delta G^{\circ}_{rxn} > 0$.
- (D) At equilibrium, for reaction 1 $\frac{[\text{Product}]}{[\text{Reactant}]} \gg 1$ and $\Delta G^{\circ}_{rxn} < 0$ but for reaction 2 $\frac{[\text{Product}]}{[\text{Reactant}]} = 0$ and $\Delta G^{\circ}_{rxn} \ll 0$.



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At 720°C , the reaction represented above has an equilibrium constant of 1.0×10^{-30} . Which of the following mathematical expressions can be used to support the claim that the reaction is not thermodynamically favored under these conditions?

(A) $\Delta G^\circ_{\text{rxn}} = \frac{-8.314 \times (720 + 273) \times \ln(1.0 \times 10^{-30})}{1,000} > 0$ ✓

(B) $\Delta G^\circ_{\text{rxn}} = -0.08206 \times (720 + 273) \times \ln(1.0 \times 10^{-30}) \times 1,000 > 0$

(C) $\Delta G^\circ_{\text{rxn}} = \frac{8.314 \times 720 \times \ln(1.0 \times 10^{-30})}{1,000} < 0$

(D) $\Delta G^\circ_{\text{rxn}} = 0.08206 \times (720 + 273) \times \ln(1.0 \times 10^{-30}) \times 1,000 < 0$



The chemical equation and thermodynamic data for the melting of tungsten are given above. Based on this information, which of the following provides the best prediction with correct justification about whether a sample of pure tungsten will melt at 3723 K ?

(A) The sample will not melt because $T > \frac{\Delta H^\circ_{\text{fus}}}{\Delta S^\circ_{\text{fus}}}$.

(B) The sample will not melt because $T < \frac{\Delta H^\circ_{\text{fus}}}{\Delta S^\circ_{\text{fus}}}$.

(C) The sample will melt because $T > \frac{\Delta H^\circ_{\text{fus}}}{\Delta S^\circ_{\text{fus}}}$. ✓

(D) The sample will melt because $T < \frac{\Delta H^\circ_{\text{fus}}}{\Delta S^\circ_{\text{fus}}}$.

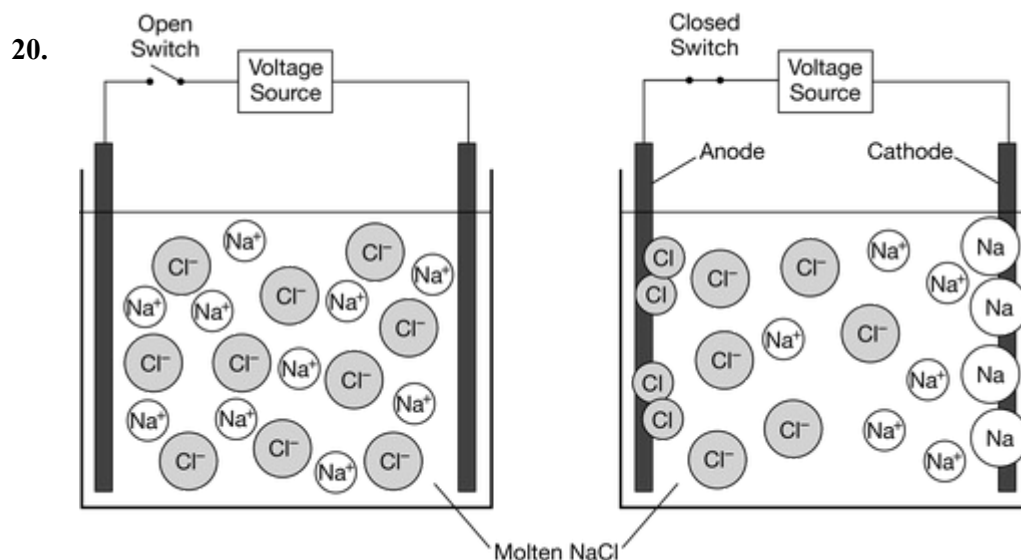


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19.	Reaction 1:	$\text{FeO}(s) + \text{CO}(g) \rightarrow \text{Fe}(l) + \text{CO}_2(g)$	$\Delta G^\circ_{rxn} > 0$
	Reaction 2:	$\text{C}(s) + \text{CO}_2(g) \rightarrow 2 \text{CO}(g)$	$\Delta G^\circ_{rxn} < 0$
	Overall reaction:	$\text{FeO}(s) + \text{C}(s) \rightarrow \text{Fe}(l) + \text{CO}(g)$	$\Delta G^\circ_{rxn} < 0$

The chemical equations above represent the main reactions that occur during the production of $\text{Fe}(l)$ under certain conditions. The overall reaction couples reactions 1 and 2, resulting in a thermodynamically favorable process. Which of the following best explains whether or not a particle diagram could represent how the coupling of reaction 1 and reaction 2 results in $\Delta G^\circ_{rxn} < 0$?

- (A) A particle diagram that represents the increase in the volume of gaseous product particles would be a good representation of how the coupling of reactions 1 and 2 results in a thermodynamically favorable process.
- (B) A particle diagram that represents the decrease in the average kinetic energy of the particles would be a good representation of how the coupling of reactions 1 and 2 results in a thermodynamically favorable process.
- (C) A particle diagram cannot represent how the changes in energy that take place as reaction 1 occurs are more than offset by the changes in energy taking place as reaction 2 occurs, resulting in a thermodynamically favorable overall reaction. ✓
- (D) A particle diagram cannot represent the changes in the amount of matter that take place as reaction 1 is coupled to reaction 2, resulting in a thermodynamically favorable overall reaction.



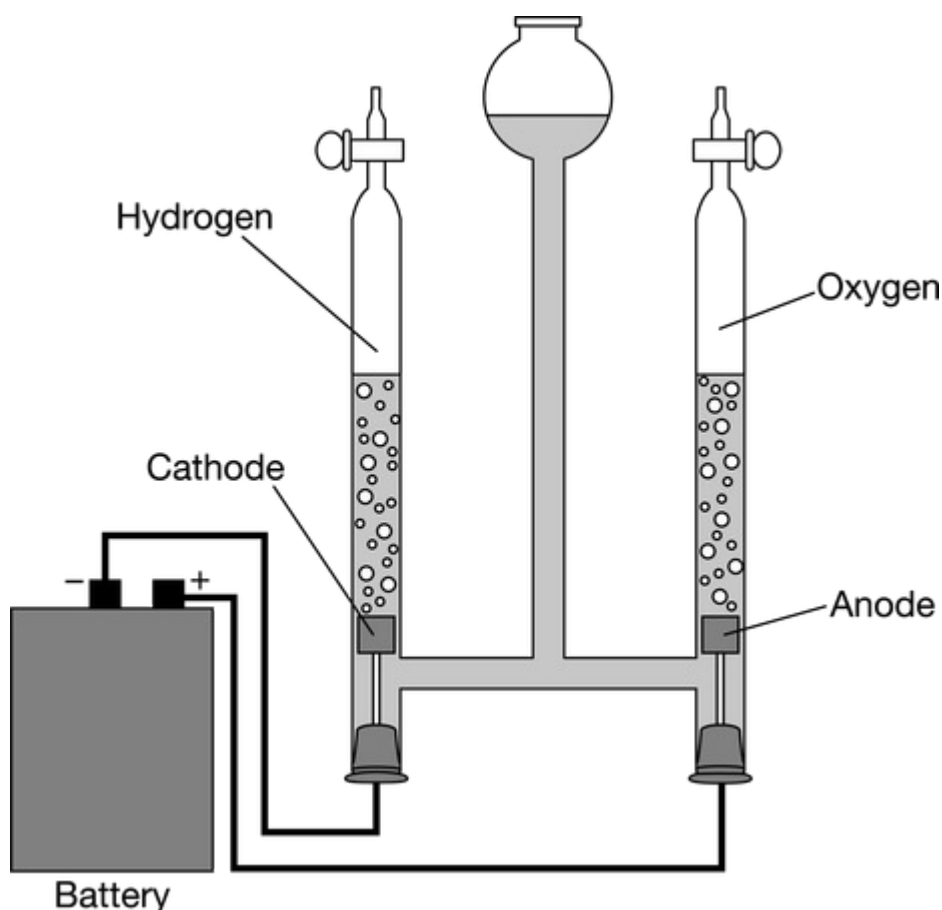
The particle diagrams above represent $\text{NaCl}(l)$ and its decomposition into $\text{Na}(l)$ and $\text{Cl}_2(g)$ in an electrochemical cell after voltage is applied. Which of the following statements about the thermodynamic favorability of the decomposition of $\text{NaCl}(l)$ is supported by the particle diagrams and why?



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- (A) The decomposition is thermodynamically favored because the transfer of electrons to and from the ions occurs at the electrodes.
- (B) The decomposition is thermodynamically favored because the formation of a gaseous product results in an increase in entropy.
- (C) The decomposition is not thermodynamically favored because the pure elements form only after electrical energy is supplied. ✓
- (D) The decomposition is not thermodynamically favored because there is a decrease in the number of particles as the reaction proceeds.

21.



The reaction in which $\text{H}_2\text{O}(l)$ is decomposed into $\text{H}_2(g)$ and $\text{O}_2(g)$ is thermodynamically unfavorable ($\Delta G^\circ > 0$). However, an electrolytic cell, such as the one represented above, can be used to make the reaction occur. Which of the following identifies a flaw in the representation?



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- (A) Oxidation is occurring at the anode.
- (B) The equation is not correctly balanced.
- (C) Electrical energy is needed for the reaction to proceed.
- (D) The volumes of collected gas in each tube should not be the same. ✓

22.

Half-Reaction	Standard Reduction Potential, E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$	+0.34
$\text{Pb}^{2+}(aq) + 2e^- \rightarrow \text{Pb}(s)$	-0.13
$\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$	-1.66

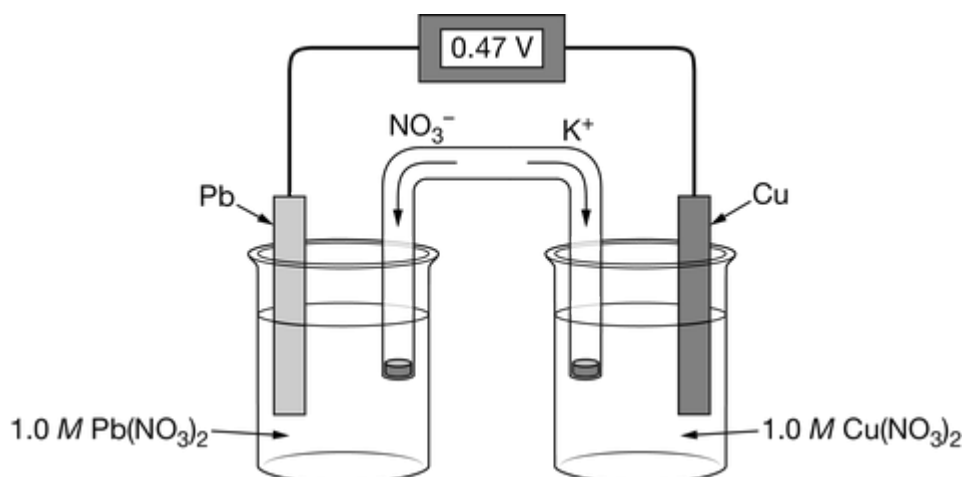
A standard galvanic cell is made using $\text{Pb-Pb(NO}_3)_2$ and $\text{Cu-Cu(NO}_3)_2$ half-cells. Which of the following modifications to the cell will cause the greatest increase in E°_{cell} , and why? Assume all solutions are 1 M.

- (A) Replacing the $\text{Cu-Cu(NO}_3)_2$ half-cell with a Ag-AgNO_3 half-cell, because the reduction of 1 mol of Ag^+ requires only 1 mol of e^- , resulting in $E^\circ_{\text{cell}} = +0.67 \text{ V}$.
- (B) Replacing the $\text{Cu-Cu(NO}_3)_2$ half-cell with a Ag-AgNO_3 half-cell, because less current flows through the cell, resulting in $E^\circ_{\text{cell}} = +0.93 \text{ V}$.
- (C) Replacing the $\text{Pb-Pb(NO}_3)_2$ half-cell with an $\text{Al-Al(NO}_3)_3$ half-cell, because the oxidation of $\text{Al}(s)$ is more thermodynamically favorable than the oxidation of $\text{Pb}(s)$, resulting in $E^\circ_{\text{cell}} = +2.00 \text{ V}$. ✓
- (D) Replacing the $\text{Pb-Pb(NO}_3)_2$ half-cell with an $\text{Al-Al(NO}_3)_3$ half-cell, because 6 mol of e^- flow through the cell, resulting in $E^\circ_{\text{cell}} = +5.66 \text{ V}$.



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23.



The galvanic cell illustrated above was constructed using a salt bridge containing KNO_3 . A second cell is constructed from identical half-cells but uses NaNO_3 for the salt bridge. Which of the following best explains whether the initial potential of the second cell will be different from the initial potential of the first cell?

- (A) The initial potential of the second cell will be lower than the initial potential of the first cell because **Na** has a lower molar mass than **K** has.
- (B) The initial potential of the second cell will be higher than the initial potential of the first cell because **Na** is less electronegative than **K** is.
- (C) The initial potential of the second cell will be the same as the first cell because the ions from the salt bridge are not oxidized or reduced during cell operation. ✓
- (D) The initial potential of the second cell will be the same as the first cell because the standard reduction potentials of Group 1 metals are very similar.

24. $2 \text{H}_2(g) + \text{O}_2(g) \rightarrow 2 \text{H}_2\text{O}(l)$ $\Delta G^\circ_{\text{rxn}} = -474 \text{ kJ/mol}_{\text{rxn}}$

Electrode	Half-Reaction	Standard Reduction Potential (V)
Cathode	$\text{O}_2(g) + 4 \text{H}^+(aq) + 4 e^- \rightarrow 2 \text{H}_2\text{O}(l)$	?
Anode	$2 \text{H}^+(aq) + 2 e^- \rightarrow \text{H}_2(g)$	0.00

The operation of a hydrogen fuel cell under standard conditions relies on the chemical reaction represented above. The table provides the relevant reduction half-reactions and the standard reduction potentials. Based on the information given, which of the following equations can be used to calculate the standard reduction potential, in volts, of the half-reaction occurring at the cathode?



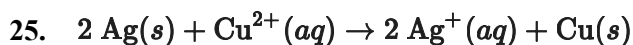
Unit 9 Progress Check: MCQ

(A) $E^\circ_{red}(\text{cathode}) = \left(\frac{-474}{4 \times 96,500} \right)$

(B) $E^\circ_{red}(\text{cathode}) = -\left(\frac{-474,000}{4 \times 96,500} \right)$ ✓

(C) $E^\circ_{red}(\text{cathode}) = \left(\frac{4 \times 96,500}{474} \right)$

(D) $E^\circ_{red}(\text{cathode}) = -\left(\frac{4 \times 96,500}{-474,000} \right)$



Half-Reaction	Standard Reduction Potential, E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Cu}^{2+}(aq) + 2 e^- \rightarrow \text{Cu}(s)$	+0.34

The reaction represented by the equation above serves as the basis for the construction of an electrochemical cell. The table gives the reduction half-reactions and their respective standard reduction potentials. Based on the overall reaction, which type of electrochemical cell was constructed, and why?

(A) A galvanic or voltaic cell, because the standard reduction potentials for the two half-reactions are positive and $E^\circ_{cell} = 0.80 + 0.34 = +1.14 \text{ V}$.

(B) A galvanic or voltaic cell, because the standard reduction potentials for the two half-reactions are positive and $E^\circ_{cell} = 0.80 - 0.34 = +0.46 \text{ V}$.

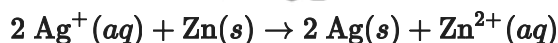
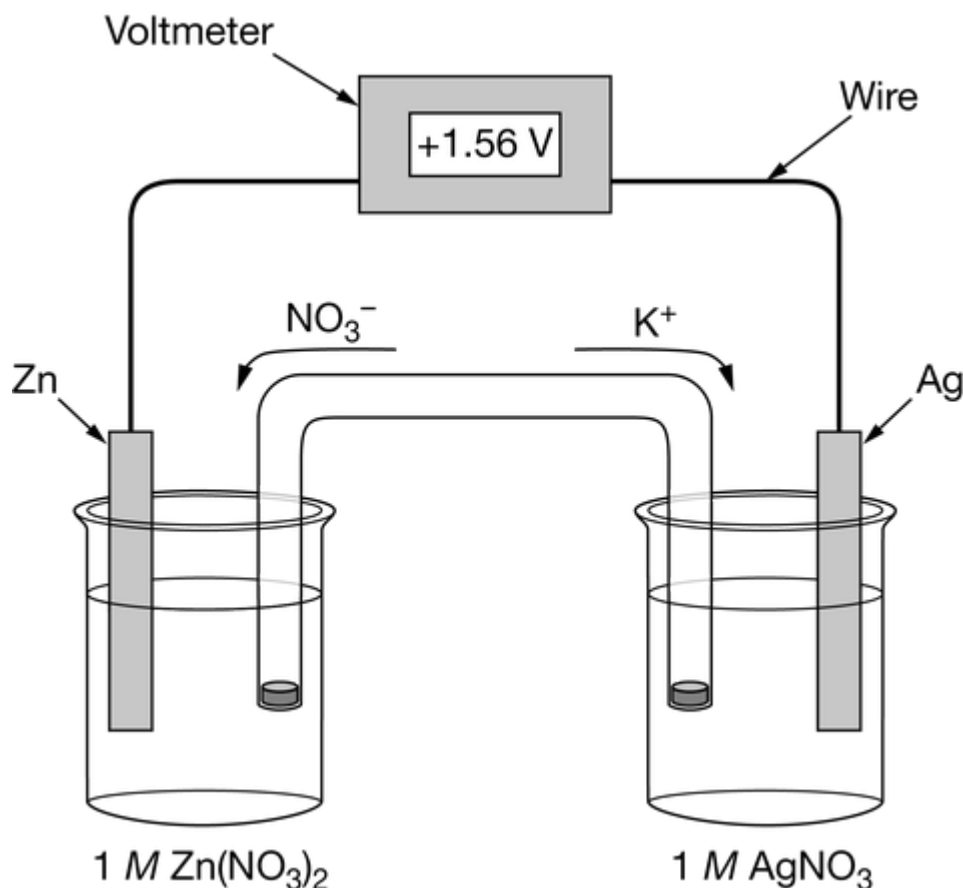
(C) An electrolytic cell, because $\Delta G = -\frac{96,500 \times (-0.46)}{1,000} = 44 \text{ kJ/mol}_{rxn}$ and its operation requires a potential smaller than 0.46 V to be supplied.

(D) An electrolytic cell, because $\Delta G = -\frac{2 \times 96,500 \times (-0.46)}{1,000} = 89 \text{ kJ/mol}_{rxn}$ and its operation requires a potential greater than 0.46 V to be supplied. ✓



Unit 9 Progress Check: MCQ

26.

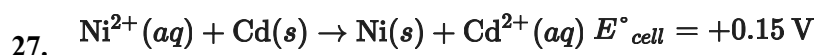


Under standard conditions, the galvanic cell shown above has a cell potential of +1.56 V using the reaction given. The salt bridge contains KNO₃, which allows K⁺ ions and NO₃⁻ ions to move in the directions indicated. If KNO₃ in the salt bridge is replaced with KOH, some Zn(OH)₂(s) precipitates in the Zn-Zn(NO₃)₂ half-cell. Which of the following best explains how the cell potential is affected as Zn(OH)₂(s) starts to precipitate, and why?

- (A) The cell potential increases because the concentration of Zn²⁺(aq) decreases and Q , $\frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$, becomes smaller. ✓
- (B) The cell potential decreases because the concentration of Zn²⁺(aq) decreases and Q , $\frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$, becomes smaller.
- (C) The cell potential increases because K⁺ ions replace Zn²⁺ ions and the reduction of K⁺ is more thermodynamically favored than the reduction of Zn²⁺.
- (D) The cell potential stays the same because Zn(OH)₂(s) is not part of the redox reaction responsible for the operation of the galvanic cell.



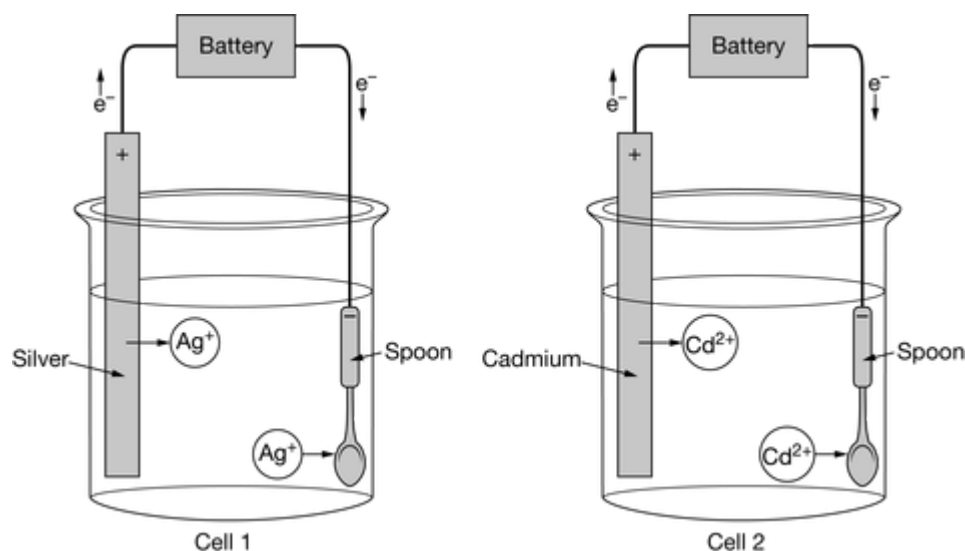
Unit 9 Progress Check: MCQ



Under standard conditions, a Ni-Cd galvanic cell generates a potential of $+0.15 \text{ V}$ using the chemical reaction represented above. Which of the following best explains how $[\text{Ni}^{2+}] > 1 \text{ M}$ affects the cell potential if this is the only change made to the operation of the galvanic cell?

- (A) The ratio $\frac{[\text{Cd}^{2+}]}{[\text{Ni}^{2+}]}$ is smaller if $[\text{Ni}^{2+}] > 1 \text{ M}$, resulting in $Q < 1$ and $E_{\text{cell}} > E^\circ_{\text{cell}}$. ✓
- (B) The ratio $\frac{[\text{Cd}^{2+}]}{[\text{Ni}^{2+}]}$ is smaller if $[\text{Ni}^{2+}] > 1 \text{ M}$, resulting in $Q > 1$ and $E_{\text{cell}} > E^\circ_{\text{cell}}$.
- (C) The ratio $\frac{[\text{Ni}^{2+}]}{[\text{Cd}^{2+}]}$ is larger if $[\text{Ni}^{2+}] > 1 \text{ M}$, resulting in $Q > 1$ and $E_{\text{cell}} < E^\circ_{\text{cell}}$.
- (D) The ratio $\frac{[\text{Ni}^{2+}]}{[\text{Cd}^{2+}]}$ is larger if $[\text{Ni}^{2+}] > 1 \text{ M}$, resulting in $Q < 1$ and $E_{\text{cell}} < E^\circ_{\text{cell}}$.

28.



Two identical spoons were electroplated with **Ag** or **Cd** through the use of the electrolytic cells illustrated above. A current of 5.00 A was supplied to each cell for $600.$ seconds, and the masses of the spoons before and after the electroplating were recorded. Which of the following mathematical equations can best be used to account for the much larger increase in mass of the spoon electroplated with **Ag** compared with the spoon electroplated with **Cd**?



Unit 9 Progress Check: MCQ

(A) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

(B) $\Delta G^\circ = -RT \ln K$

(C) $E = E^\circ - \left(\frac{RT}{nF}\right) \ln Q$

(D) $I = \frac{q}{t}$



29.

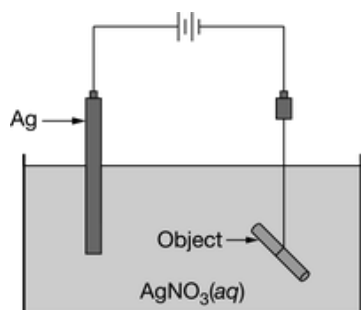


Diagram 1

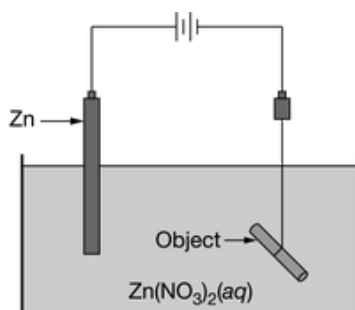


Diagram 2

Object	Metal Used	Amount of Metal Deposited (mol)	Time of Operation (minutes)
1	Ag	0.10	30
2	Ag	0.10	60
3	Zn	0.10	30
4	Zn	0.10	60

The diagrams above illustrate the equipment used to electroplate four identical objects with silver or zinc. The table provides the conditions used. Which of the following provides the basis for the identification of the object that requires the highest current, I , to complete the electroplating?



Unit 9 Progress Check: MCQ

- (A) Object 1, based on the molar mass of **Ag** and the time of operation.
- (B) Object 2, based on the molar mass of **Ag** and the time of operation.
- (C) Object 3, based on the coulombs of charge needed and the time of operation. ✓
- (D) Object 4, based on the coulombs of charge needed and the time of operation.

30.

Half-Reaction	Standard Reduction Potential, E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Au}^{3+}(aq) + 3 e^- \rightarrow \text{Au}(s)$	+1.50

Two different electrolytic cells are constructed to electroplate two identical objects based on the half-reactions given in the table above. An object will be plated with a certain number of moles of **Au** in one cell, and in the other cell another object will be plated with the same number of moles of **Ag** in the same amount of time. Which of the following mathematical relationships can be used to determine how much more current is needed to electroplate an object with **Au** than to electroplate it with **Ag**?

- (A) The difference in standard reduction potential of Au^{3+} and Ag^+ is 0.70 V.
- (B) Three times more electrons are needed to reduce $\text{Au}^{3+}(aq)$ to $\text{Au}(s)$ than are needed to reduce $\text{Ag}^+(aq)$ to $\text{Ag}(s)$. ✓
- (C) ΔG° is more negative for the reduction of $\text{Au}^{3+}(aq)$ to $\text{Au}(s)$ than for the reduction of $\text{Ag}^+(aq)$ to $\text{Ag}(s)$.
- (D) The molar-mass ratio of $\frac{\text{Au}}{\text{Ag}}$ is $\frac{196.97}{107.87}$.